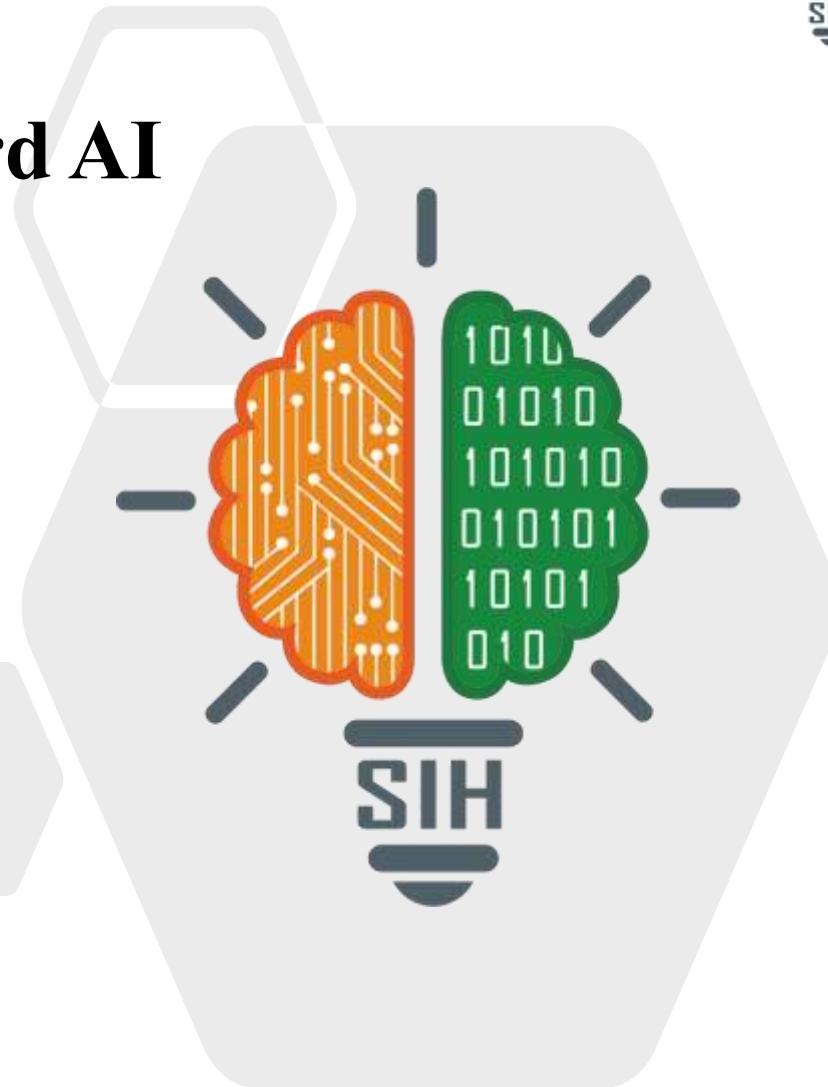


CampusGuard AI

- Problem Statement ID –
SIH2026_PS_4609
- Problem Statement Title- AI-Based
Real-Time Campus Security &
Surveillance System
- Theme- Smart Automation

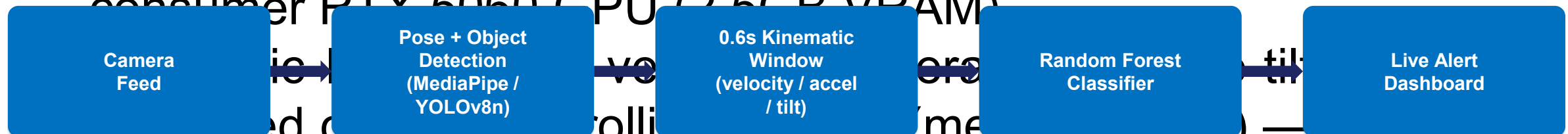


❖ Proposed Solution (Describe your Idea/Solution/Prototype)

- A real-time, privacy-first computer vision pipeline that acts as an automated digital watchman — running 100% on local edge hardware (e.g. a consumer RTX 5060 GPU), simultaneously detecting fights, falls and unattended bags, with live alerts on a security dashboard
- Replaces manual monitoring of dozens of CCTV feeds — eliminates blind spots, cuts emergency response delays for fights/falls, and flags unattended bags automatically, without ever sending video to the cloud (fully compliant with student data-privacy laws)

TECHNICAL APPROACH

- Vision: MediaPipe PoseLandmarker (Heavy) — 33 keypoints, up to 8 people/frame; Ultralytics YOLOv8n tuned on COCO bag/person classes | ML: Scikit-Learn Random Forest (fight/fall) | Runs 100% on-device on a consumer RTX 5060 CPU (2.5GB VRAM)



sustained-threshold alerts, not single-frame guesses.
Train/Serve Parity: batch_extractor.py and app.py share identical math, eliminating train/serve skew

FEASIBILITY AND VIABILITY

- Optimized to run on a consumer-grade RTX 5060 (2.5GB VRAM) using a 2-frame skip protocol for an effective 5 FPS — sufficient for the 0.6s kinematic window to reliably catch violent action

2.5 GB

VRAM on consumer
RTX 5060 GPU

5 FPS

Effective rate via
2-frame skip protocol

83%

Recall — Fight
Detection (F1: 0.739)

85%

Recall — Fall
Detection (F1: 0.685)

data leakage between train/test, windowed statistical
thresholds curb false positives, and the modular

IMPACT AND BENEFITS



- Direct impact for students, faculty and security teams across every camera-covered campus zone — turning passive CCTV footage into an active, always-on safety net

SOCIAL

Faster emergency response to fights & falls; eliminates CCTV blind spots across campus

ECONOMIC

Scalable alternative to costly round-the-clock human monitoring of large camera networks

PRIVACY

Zero-leakage local processing — no video ever leaves campus, ensuring full data-law compliance

- Privacy-by-design: inference runs entirely on-device on a consumer RTX 5060; no video or pose data is uploaded off-campus, satisfying student-data privacy requirements.

Datasets

RWF-2000 — 1,153 real-world fight clips
UR Fall Dataset — RGB-D fall videos
CAUCAFall — multimodal fall kinematics

Models

Google MediaPipe PoseLandmarker
Ultralytics YOLOv8n (COCO classes)
Scikit-Learn Random Forest

Method

0.6s rolling-window kinematics
Train/serve parity (batch_extractor == app.py)
GroupShuffleSplit @ video level

Links

[Add: GitHub repo link / demo video link]

- Performance: Fight Macro-F1 0.739 / Recall 83%; Fall Macro-F1 0.685 / Recall 85%, ~2.5 GB VRAM at ~5